

Project - Tech Research

Learning Objectives

- Conduct independent research on an emerging technology topic
- Synthesize information from multiple authoritative sources
- Present technology analysis with balanced technical and ethical perspectives

Engage (5-7 minutes)

Show a montage of emerging technologies: CRISPR gene editing, quantum computing, brain-computer interfaces, fusion energy, and space tourism. Ask students: which of these will most transform your lifetime? Students vote on their top three, and the class forms research groups around the most popular topics.

Explore (10-12 minutes)

Each group selects (or is assigned) a technology topic from: AI ethics, IoT security, blockchain for social good, 5G and digital divide, cloud sustainability, or quantum computing prospects. They develop a research plan: key questions, credible sources (academic papers, industry reports, reputable journalism), interview contacts, and a timeline. Groups conduct research over multiple sessions, documenting findings and cross-referencing sources.

Explain (10-12 minutes)

The research project synthesizes Chapter 6 concepts: students must explain the technology's technical foundations (how it works), current applications (what it does now), future potential (what it could do), ethical considerations (what could go wrong), and societal impact (who benefits and who is harmed). The final deliverable includes: a 10-minute presentation, a visual poster or infographic, a bibliography of at least 8 sources, and a one-page policy recommendation. Research skills emphasized: source evaluation, bias detection, and distinguishing fact from opinion.

Elaborate (8-10 minutes)

Students develop professional communication skills: presenting complex technical concepts to non-specialist audiences, handling Q&A, and defending positions with evidence. They practice peer feedback using structured rubrics. Academic integrity is reinforced: proper citation, paraphrasing versus copying, and the importance of original analysis. Groups revise their work based on feedback before final presentation.

Evaluate (5-8 minutes)

Final assessment combines: presentation quality (technical accuracy, clarity, engagement), research depth (source quality, breadth of coverage, critical analysis), written report (structure, argumentation, citations), and peer evaluation (collaboration, contribution, constructive feedback). Self-assessment: students reflect on their research process, what they learned about the technology, and what they would investigate further.